

Free Data Sources

Jan Masner



Free data sources

- (National) statistical offices
 - <https://www.czso.cz/csu/czso/home>
 - <https://ec.europa.eu/eurostat>
- Organizations
 - <https://data.europa.eu/en>
 - <https://data.worldbank.org>
 - <https://www.who.int/data>

(National) statistical offices

Český statistický úřad (ČSÚ) / Czech Statistical Office (CZSO)

Statistics:

- *People* (population, foreigners etc.)
- *Society* (elections, education, health, justice etc.)
- *Regions and Countries* (summary data on Czechia, regional statistics, territory, international data etc.)
- *Economy* (prices, inflation, international trade, GDP etc.)
- *Economic sectors* (industry, energy, transport, tourism, trade, services, agriculture etc.)
- *Labour market* (employees and wages, labour costs, structure of earnings etc.)
- *Science, innovation and digitalization* (research and development (R&D), innovation, patents, high technology, digital society - ICT usage, digital economy etc.)

Eurostat (European Statistical Office)

Statistical themes

- EU policy indicators (Euro indicators, Sustainable development, European pillar of social rights etc.)
- General and regional statistics (Regions and cities, Land cover and use, International cooperation)
- Economy and finance (National accounts (including GDP), ESA supply, use and input-output tables, Exchange and interest rates, Housing price statistics)
- Population and social conditions (Population and demography, Migration and asylum, Health, Disability, Education, Income and living conditions etc.)
- Industry, trade, services (Short-term business statistics (STS), Structural business statistics (SBS), Business demography, Globalisation of businesses)
- Agriculture, fisheries, and forestry
- International trade (in goods and in services)
- Transport
- Environment and energy (Energy, Environment, Waste, Climate change)
- Science, technology, and digital society

Eurostat (European Statistical Office)

Databases:

- General and regional statistics
- Economy and finance
- Population and social conditions
- Industry, trade and services
- Agriculture, forestry and fisheries
- International trade
- Transport
- Science, technology, digital society
- EU policies etc.

Organizations

data.europa.eu (European Data Portal)

The information is structured into 1,947,909 datasets, 204 catalogs and editorial content, covering 35 countries

- European Union Data
- International Data
- National Data

data.europa.eu (European Data Portal)

Formats:

- WMS, WFS
- CSV, JSON, HTML, XML
- PDF
- ZIP
- Excel XLSX, XLS etc.

data.europa.eu (European Data Portal)

Categories:

- Agriculture, fisheries, forestry and food
- Justice, legal system and public safety
- Environment
- Government and public sector
- Economy and finance
- Science and technology
- Population and society
- Regions and cities
- Transport
- Education, culture and sport
- Health
- Energy
- Provisional data
- International issues

World Bank Open Data (data.worldbank.org)

The Community section is a space for connecting with other data enthusiasts, professionals, and stakeholders.

- Apps
- Podcasts
- Collaboration channel
- Events

WHO Data (World Health Organization – Data Platform)

Global Health Observatory:

- Air pollution
- Antimicrobial resistance (AMR)
- Assistive technology
- Environment and health
- Global Health Estimates
- Global patient safety
- Health financing
- Health workforce
- Maternal and reproductive health
- Mental health
- Water, sanitation and hygiene (WASH)
- Etc.

WHO Data (World Health Organization – Data Platform)

Featured portals:

- Maternal, Newborn, Child and Adolescent Health and Ageing
- Global Strategy for Women's, Children's and Adolescents' Health (2016-2030)
- Global Observatory on Health R&D
- Sexual and Reproductive Health and Rights Policy Portal

WHO Data (World Health Organization – Data Platform)

Featured data sets:

- Global health estimates
- Mortality database
- Health SDGs
- Immunization data
- Mortality due to pregnancies with abortive outcome

Other data sources

- List of open data repositories/sources
 - https://oad.simmons.edu/oadwiki/Data_repositories
 - <https://github.com/awesomedata/awesome-public-datasets>
 - <https://data.world/datasets/free>
 - <https://dataportals.org>
 - <https://mavenanalytics.io/data-playground>
 -
- Google data
 - <https://www.google.com/publicdata/directory>
 - <https://www.google.com/finance/>
 - <https://trends.google.com/home?hl=en-US>
- Inspiration/data
 - <https://ourworldindata.org>
 - <https://www.statista.com>

List of open data repositories/sources

Open Access Directory – Data Repositories

This is a list of repositories and databases for open data:

- Archaeology (also see Social sciences)
- Astronomy (also see Physics)
- Biology
- Chemistry
- Computer science (2 million documents and the associated texts and pdfs)
- Energy
- Engineering (also see Multidisciplinary repositories)
- Environmental sciences

Open Access Directory – Data Repositories

This is a list of repositories and databases for open data:

- Geology
- Geosciences and geospatial data
- Linguistics (see the 40+ members of the Open Language Archives Community)
- Marine sciences
- Medicine
- Multidisciplinary repositories (also see Social Sciences)
- Physics (also see Astronomy)
- Social sciences (also see Multidisciplinary repositories)

Awesome Public Datasets

([awesomedata/awesome-public-datasets](https://awesomedata.github.io/awesome-public-datasets/))

This is a list of topic-centric public data sources in high quality:

- Agriculture
- Architecture
- Biology
- Chemistry
- Climate & Weather
- Complex Networks
- Computer Networks
- Cyber Security
- Data Challenges
- Earth Science
- Economics
- Education

Awesome Public Datasets

([awesomedata/awesome-public-datasets](https://awesomedata.com/awesome-public-datasets))

This is a list of topic-centric public data sources in high quality:

- Energy
- Entertainment
- Finance
- GIS
- Government
- Healthcare
- Image Processing
- Machine Learning
- Museums
- Natural Language
- Neuroscience
- Physics

Awesome Public Datasets

([awesomedata/awesome-public-datasets](https://awesomedata.com/awesome-public-datasets))

This is a list of topic-centric public data sources in high quality:

- Psychology & Cognition
- Public Domains
- Search Engines
- Social Networks
- Social Sciences
- Software
- Sports
- Time Series
- Transportation
- eSports
- Complementary Collections

data.world Free Datasets

It is a collaborative open data platform where individuals, organizations, and researchers can share, explore, and analyze datasets. It combines features of a social network and a data catalog, supporting data discovery, documentation, SQL querying, and integration with analytical tools.

Main features:

- Upload, publish, and share datasets (public or private)
- Browse thousands of open/free datasets contributed by users and institutions
- Discuss, comment, and track dataset activity
- Query data directly online using SQL or SPARQL
- Integrate with tools like Tableau, Python, R, and BI platforms
- Access data via API or download as CSV/ZIP

data.world Free Datasets

The platform offers datasets across a wide range of domains, such as:

- Public health and COVID-19 data
- Environment, climate, and sustainability
- Education, demographics, and economics
- Social and government statistics
- Research and scientific datasets

Featured examples:

- *COVID-19 Data Resource Hub* – global pandemic tracking and policy data
- *Intro Datasets & Exercises* – curated datasets for learning and data practice
- *US Government and City Open Data* – public administration and infrastructure statistics

dataportals.org

Indexes hundreds of open data portals worldwide (local, regional, national) — over 600 portals listed.

For each portal, provides metadata such as:

- Name, region/country or jurisdiction
- Link to the portal
- Tags / categories (local, national, sector, etc.)
- Portal software / API information (if available)
- Status / activity context

The catalog is community-driven: users can suggest additions or corrections via GitHub issues.

Maven Analytics Data Playground

A curated collection of real-world datasets meant for practicing data analysis, visualization, and storytelling skills — without needing to pay. These datasets are used in Maven's courses, challenges, and learning activities, making them especially suited for hands-on learning.

Wide domain variety — datasets span many topics: retail, healthcare, environment, finance, sports, transportation, surveys, etc.

Different data structures — some datasets are in a single table, others use multiple tables (relational style)

Metadata available — each dataset shows number of rows (records), number of fields (columns), tags/themes (e.g. “Time Series”, “Geospatial”)

Maven Analytics Data Playground

Example Datasets

- *Hospital Patient Records* — around 75.5k records of about 1k patients.
- *The Museum of Modern Art (MoMA) Collection* — around 157.5k records of about 150k works.
- *LEGO Sets* — about 18.5k records of LEGO sets released between 1970 and 2022.
- *Bank Customer Churn* — around 10k records of account information for 10k customers at a European bank.
- *Global Electronics Retailer* — around 62.8k records of transactional data for a fictitious global electronics retailer.

Google data

datacommons.org

The United Nations deployed its own Data Commons to track their Sustainable Development Goals data around the world. With this, the UN is able to offer a centralized repository, allowing for dynamic storytelling and targeted analysis related to global progress.

Data Commons tools:

Data Commons addresses offers data exploration tools and cloud-based APIs to access and integrate cleaned datasets.

- Map Explorer
- Scatter Plot Explorer
- Timelines Explorer
- Data Download Tool
- API Access

Organizations who have built their own Data Commons:

- United Nations Sustainable Development Goals
- Data Commons for civil society (TechSoup)
- Feeding America
- Stanford University

Google Finance

It is a free platform by Google that provides **real-time financial data, market news, and investment tracking**. It allows users to monitor stocks, ETFs, indices, and cryptocurrencies, as well as analyze market trends and build personalized portfolios.

Google Finance Tools:

- Real-Time Stock Quotes
- Interactive Charts (candlestick, technical indicators)
- News Integration (curated by company/market events)
- Portfolio Tracking
- Google Sheets Integration (GOOGLEFINANCE function)

Google Trends

It is a free online tool by Google that shows **how frequently specific search terms are entered across various regions and languages over time**. It allows users to explore search interest patterns, compare topics, and uncover trends in real time or historically. Google Trends provides insights into public interest and behavior, making it a valuable tool for marketers, researchers, journalists, and anyone studying social and cultural dynamics.

Google Trends

Google Trends Tools and Features:

- **Search Term Comparison** – compare popularity of multiple terms over time
- **Regional Interest Maps** – visualize search interest geographically
- **Trending Searches** – view currently popular searches globally or by country
- **Category Filtering** – narrow searches by topic categories (e.g., Business, Health, Entertainment)
- **Time Range Filtering** – analyze trends from real-time data to decades of historical data
- **Related Queries and Topics** – discover associated searches and rising trends
- **Data Download** – export search interest data as CSV for further analysis



Data APIs

- From pictures of cats to weather data...
- <https://github.com/public-apis/public-apis>
- <https://apilist.fun>
- <https://any-api.com>
- Via browser or, e.g., <https://www.postman.com>

From pictures of cats to weather data...

github.com public-apis

The Public APIs repository is manually curated by community members like you and folks working at APILayer. It includes an extensive list of public APIs from many domains that you can use for your own products. Consider it a treasure trove of APIs well-managed by the community over the years.

github.com public-apis

APILayer APIs:

- *IPstack* – Locate and Identify Website Visitors by IP Address
- *Marketstack* – Free, easy-to-use REST API interface delivering worldwide stock market data in JSON format
- *Weatherstack* – Retrieve instant, accurate weather information for any location in the world in lightweight JSON format
- *Numverify* – Global Phone Number Validation & Lookup JSON API
- *Fixer* – Fixer is a simple and lightweight API for current and historical foreign exchange (forex) rates.
- *Aviationstack* – Free, real-time flight status and global Aviation data API

apist.fun

It is a community-curated directory offering a wide range of **free and fun APIs** for developers. It provides an accessible platform to discover APIs across various categories, including entertainment, education, and utilities, making it ideal for both learning and integrating into projects.

Educational Projects: Utilize APIs for learning purposes and building educational tools.

Prototype Development: Quickly integrate APIs into prototypes for testing and demonstration.

Entertainment Applications: Incorporate fun APIs to enhance user engagement in apps and websites.

Data Analysis: Access various data APIs for analysis and visualization projects.



APIs are organized into a wide variety of categories, covering hundreds of topics from Animals and Music to Games, ECommerce, Education, Food, Science, Religion and more.

The best APIs:

- Randomfox (Random pictures of foxes)
- Spotify Web (Let's you search and lookup metadata about artists, tracks and albums)
- YouTube API (Allows developers to access video statistics and YouTube channels' data via two types of calls, REST and XML-RPC)

APILayer

APILayer Marketplace is a curated platform offering a wide range of **reliable and scalable APIs** for developers and businesses. It provides access to over 75 APIs across various categories, including finance, geolocation, communication, and more. Each API comes with a free plan, making it accessible for both small projects and enterprise applications.

APILayer

Extensive API Collection: Access to APIs in categories such as currency exchange, number verification, weather data, and more.

Unified Authentication: Single API key for all services, simplifying integration.

Detailed Documentation: Comprehensive guides and examples to facilitate easy implementation.

Flexible Pricing Plans: Options ranging from free tiers to enterprise solutions, catering to various needs.

Developer Support: Access to a community of over 450,000 developers, providing insights and assistance.



APILayer

Financial Applications: Integrate real-time exchange rates and currency conversion features.

Communication Platforms: Utilize number verification and email validation APIs to enhance user registration processes.

Geolocation Services: Implement IP geolocation for personalized content delivery.

Data Analytics: Leverage weather and market data APIs for analytical insights.

APILayer

Popular APIs:

- **Ipstack** – Locate and Identify Website Visitors by IP Address.
- **Exchange Rates Data API** – Real-Time, Intraday & Historical Market Data API.
- **Currency Data API** – Reliable Exchange Rates & Currency Conversion for your Business.
- **Number Verification API** – Global Phone Number Validation & Lookup JSON API.
- **Fixer API** – Foreign exchange rates and currency conversion JSON API.
- **Email Verification API** – Validates and verifies an email address assessing deliverability and quality.

Via browser or, e.g., postman.com

It is an all-in-one **API platform** designed to support every stage of the API lifecycle — from design and testing to monitoring and collaboration. It helps teams build, share, and manage APIs efficiently in a single, unified environment.

Developers building and testing APIs.

QA engineers performing automated testing.

Teams collaborating on API design and documentation.

Organizations enforcing governance and consistency in large API ecosystems.

Via browser or, e.g., postman.com

Postman can connect to many popular tools:

- CircleCI
- Jenkins
- GitHub/GitLab
- AWS API Gateway
- New Relic
- Datadog
- Slack and many others.